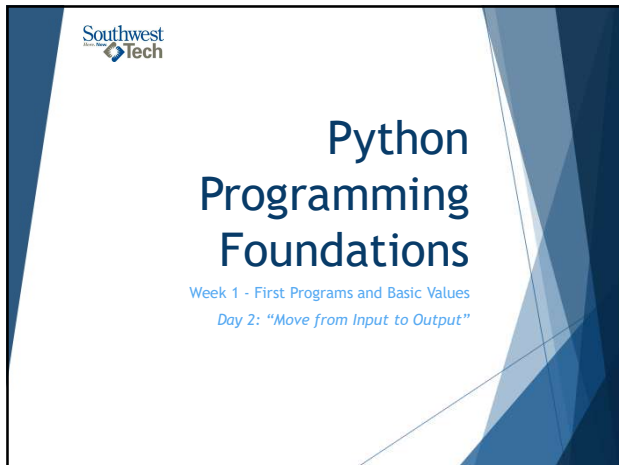
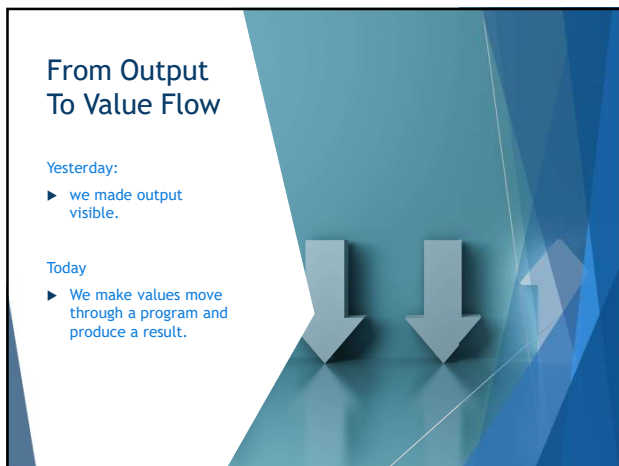
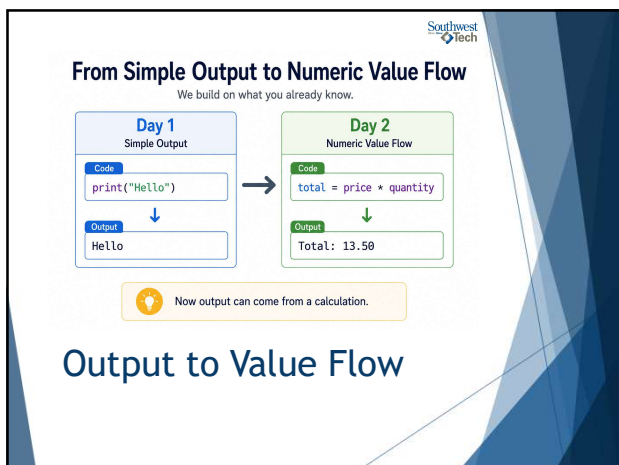




1



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Quick Review

A variable name refers to a stored value.

`'print()'` lets us see that value on the screen.



4

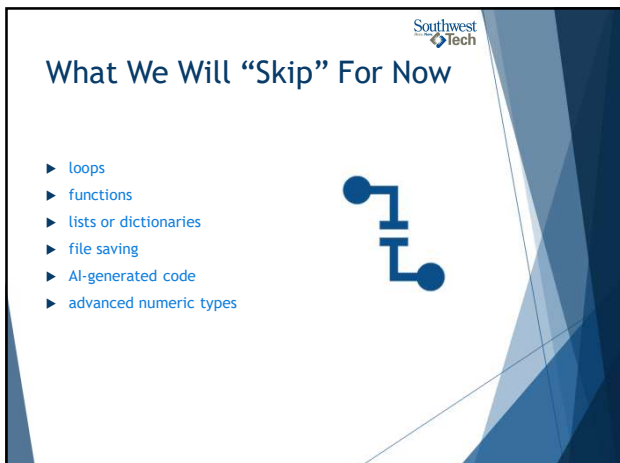


What We Will Use Today

- ▶ Numbers
- ▶ simple expressions
- ▶ result variables
- ▶ `'print()'`

`'input()'` is optional and only useful if it does not distract you from value flow.

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What We Will "Skip" For Now

- ▶ loops
- ▶ functions
- ▶ lists or dictionaries
- ▶ file saving
- ▶ AI-generated code
- ▶ advanced numeric types

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Today's Beginner Python Working Set

Master a few simple tools first.

Today (Included)
These are in your toolbox.

print()
Make Python produce visible output.

strings
Work with text and messages.

variables
Store and use information in memory.

run/run
Run your code. Set results. Try again.

Later (Coming Next)
More tools for bigger jobs.

input()

calculations

loops

functions

AI-generated code

Working Set

★ Focus today. Build confidence. The rest comes next.

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Numbers Are Values Too

Python can store numbers just like it stores text.

Numbers can also be used in calculations.

Strings and Numbers as Stored Values
Different kinds of values. Same idea.

String Value
Text is stored as a string.

variable name → stored value

student_name → "Jordan"

This is text. We can quote it.

Number Value
Numbers are stored as numbers.

variable name → stored value

items_purchased → 3

This is a number. We can't quote it.

Both can be stored in variables. Python can calculate with numbers.

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Expressions Create New Values

Expressions Create New Values
Combine values. Get a new value.

Input Values

items_purchased = 3

price_each = 4.50

→

Expression

items_purchased * price_each

→

New Value (Result)

subtotal = 13.50

An expression uses values to create a result.

An expression combines values and produces a new value.

Example:

- 'items_purchased * price_each' creates a subtotal.

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Result Variables

- ▶ A result variable stores the answer from an expression.
- ▶ Clear names help you explain what the answer means.

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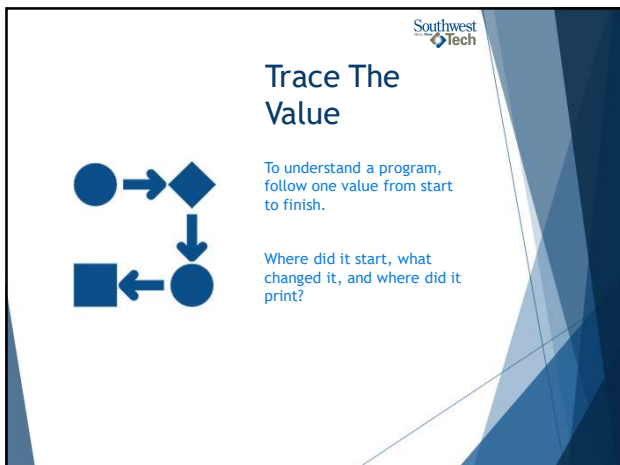
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Output Should Mean Something

Printing a number is better when the output tells us what the number means.

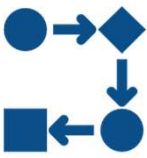
`'print("Subtotal:", subtotal)'` is clearer than printing `'subtotal'` alone.

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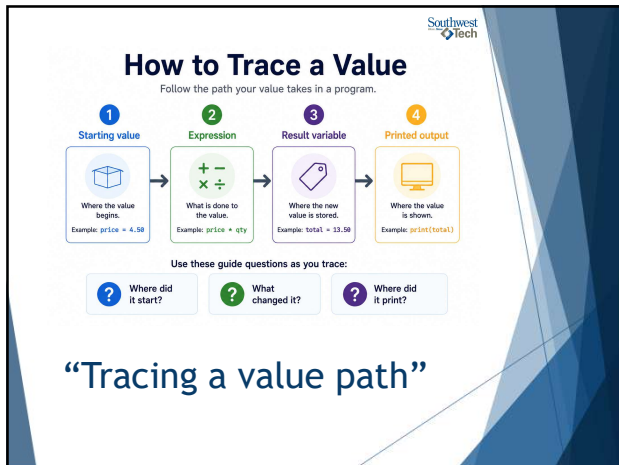
Trace The Value



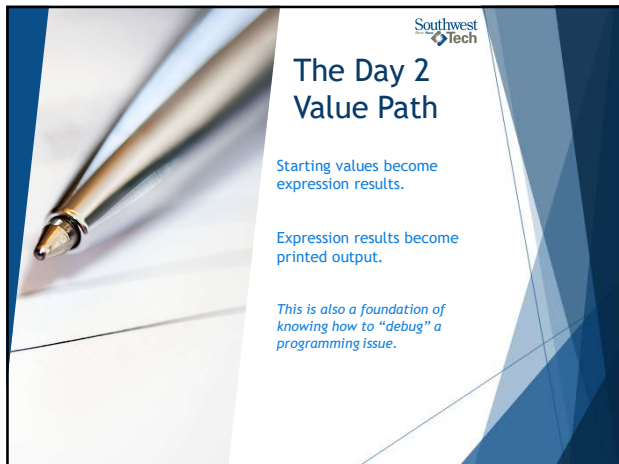
To understand a program, follow one value from start to finish.

Where did it start, what changed it, and where did it print?

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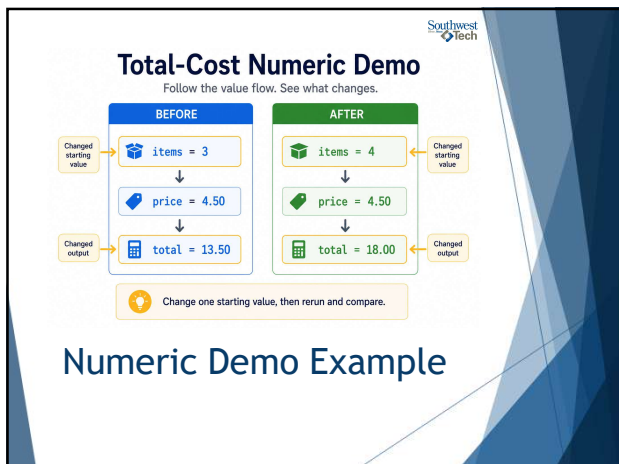


What to watch for

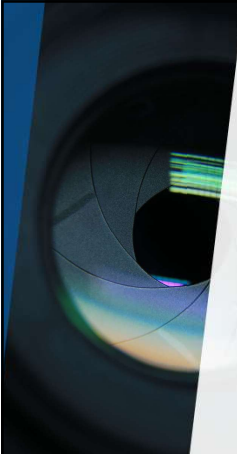
As the demo runs, watch for two things:

- ▶ which values are stored
- ▶ which lines make output appear

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Optional Demo #2

"Simple Input"

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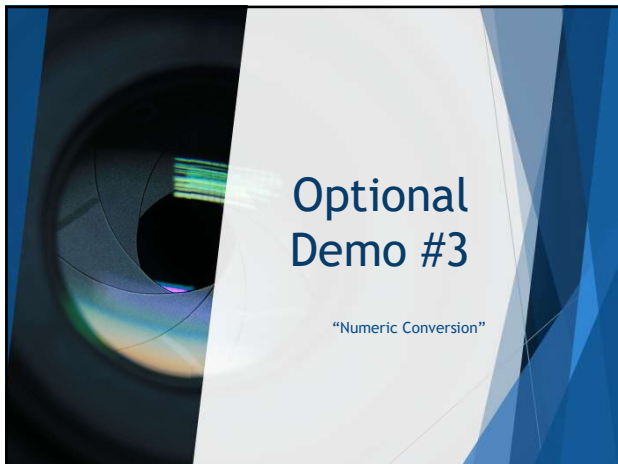


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Optional Demo: Simple `input()`

- ▶ `input()` can collect a user response.
- ▶ That response still becomes a value the program must trace.

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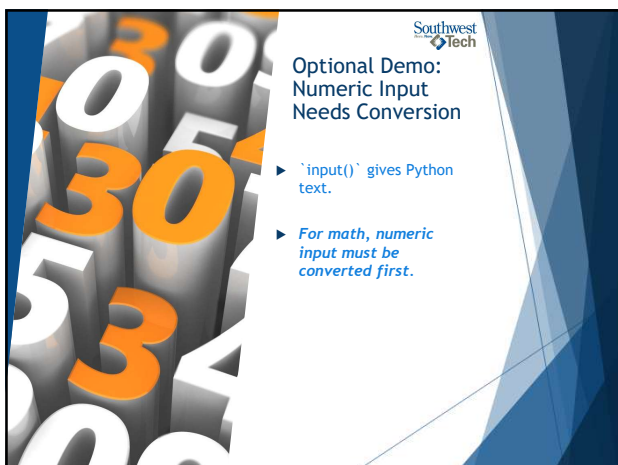


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Optional Demo #3

"Numeric Conversion"

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Optional Demo: Numeric Input Needs Conversion

- ▶ `input()` gives Python text.
- ▶ *For math, numeric input must be converted first.*

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Numeric Input: Convert Before Math

What you type is text. Convert it to a number before calculating.

```

graph LR
    A[User types: "5"] --> B[text first: input() receives text (a string). "5"]
    B --> C[convert: float("5") converts text to a number. 5.0]
    C --> D[calculate: Number can now be used in math. 5.0]
  
```

Key idea: Your keyboard input is text first. Convert it to a number, then do the math.

You may use `input()` if needed. It is *optional* for this assignment.

Numeric Conversion Example

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Common Failure: Losing The Value

- When several variables appear ... it is easy to lose the path.

Slow down and ask:

- "Where did this value come from?"

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Common Failure: ``input()`` Can Distract

- ``input()`` is useful, but it is not the main goal today.
- The main goal is understanding how a value becomes output.

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From Demo to Lab

In Lab/Assignment:

- ▶ Build one small script that starts with numbers and prints a result.

Good choices:

- ▶ Calculator
- ▶ Converter
- ▶ Total-cost estimator

This is a continuation of the first assignment.

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Assignment 1: Three Beginner Script Choices

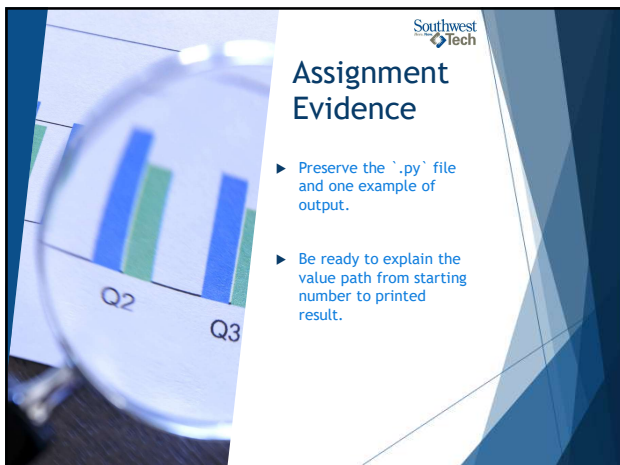
Each script uses one expression and prints one result.

1 Simple Calculator	2 Unit Converter	3 Total-Cost Estimator
Starting value num = 7	Starting value miles = 10	Starting value items = 3 price = 4.50
Expression num + 5	Expression miles * 1.609	Expression items * price
Printed result Result: 12	Printed result Kilometers: 16.09	Printed result Total: 13.50

Choose one small path and keep it traceable.

Hands-On Choices

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Assignment Evidence

- ▶ Preserve the '.py' file and one example of output.
- ▶ Be ready to explain the value path from starting number to printed result.

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Day 2 Evidence

Show what you built and what happened.

1. File



cost_estimator.py

2. Output

```
$ python cost_estimator.py
Total: 18.00
$
```

3. Note

items changed from 3 to 4, so the total changed

★ Preserve the code, output, and value path.

Assignment Evidence

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Closing Success Check

By the end of today you should be able to:

- ▶ store numbers,
- ▶ calculate a result
- ▶ print it clearly.
- ▶ You should also be able to trace where the printed result came from.

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Thank you!

Have Fun!

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